

Institute of Actuaries of India

Subject CT6 – Statistical Methods

October 2015 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:-

The costs incurred by the student attending coaching classes depending on the number of classes are:

No of classes	Cost(to and fro) from coaching class (in Rs)
1	100
2	200
3	300
4	400
5	500

If he purchases the weekly train pass, the cost incurred equals Rs 320

Let d1 = decision of not buying the weekly pass

d2 = decision of buying the weekly pass

Minimax criterion:

The maximum cost under the decision (d1) of not purchasing weekly train pass is Rs 500

The maximum cost under the decision (d2) of purchasing weekly train pass is Rs 320

The minimax solution is the minimum of the maximum costs under the two decision functions i.e. decision d2

Thus under the minimax criterion the student should choose to buy a weekly train pass

Bayes Solution:

The decision which would minimise the expected cost would be the Bayes solution to this problem.

Under d1, the expected cost equals

$$100 * .2 + 200 * .1 + 300 * .3 + 400 * .25 + 500 * .15$$

$$= 20 + 20 + 90 + 100 + 75$$

$$= 305$$

Under d2, the expected cost is the cost of purchasing the pass, which is

$$= 320$$

Thus d1 i.e. not purchasing a weekly train pass, is the decision which minimises the expected cost and hence is the Bayes Solution to this problem.

[5 Marks]

Solution 2:-

The chi-square distribution with 10 degrees of freedom is same as Gamma distribution with parameters

$$\alpha = 5, \lambda = 0.5$$

Also, Gamma distribution can be expressed as a sum of exponential distribution.

If X_i follows exponential distribution then

$$M_{X_i}(t) = \frac{\lambda}{\lambda - t}$$

Then

$$M_{\sum X_i}(t) = (M_{X_i}(t))^n = \left(\frac{\lambda}{\lambda - t}\right)^n$$

Which is the mgf of Gamma distribution with parameters n and λ .

Hence the random sample can be generated using 5 random numbers from exponential distribution (with parameter 0.5) and adding them together.

Now we have,

$$F_X(x) = 1 - e^{-0.5x} = u$$

Where u is a pseudo random number from $U(0,1)$

So random numbers from exponential distribution can be found as given below:

$$u = 0.271, x = -\ln(1-u)/0.5 = 0.632$$

$$u = 0.399, x = 1.018$$

$$u = 0.240, x = 0.549$$

$$u = 0.704, x = 2.435$$

$$u = 0.285, x = 0.671$$

Adding them above we get the random number from chi-square distribution which is 5.305.

[7 Marks]

Solution 3:-

i)

Let denote A_i and X_i as the average total claim amount per policy and total number of claims in year i .

Then X_i has a Poisson distribution with parameter (no of policy $\times \mu$).

Hence, $X_1 = (100 \times a_1)/1500$, $X_2 = (130 \times a_2)/1700$, $X_3 = (180 \times a_3)/2000$

Now we will find the posterior distribution of μ .

$$\begin{aligned} f(\mu | a_1, a_2, a_3) &\propto f(a_1, a_2, a_3 | \mu) f(\mu) \\ &\propto e^{-100\mu} (100\mu)^{\frac{100a_1}{1500}} e^{-130\mu} (130\mu)^{\frac{130a_2}{1700}} e^{-180\mu} (180\mu)^{\frac{180a_3}{2000}} e^{-\lambda\mu} \mu^{\alpha-1} \\ &\propto e^{-(100+130+180+\lambda)\mu} \mu^{(\alpha + \frac{100a_1}{1500} + \frac{130a_2}{1700} + \frac{180a_3}{2000})-1} \end{aligned}$$

Hence the posterior distribution of μ is a Gamma distribution with parameters $(\alpha + \frac{100a_1}{1500} + \frac{130a_2}{1700} + \frac{180a_3}{2000})$ and $(100 + 130 + 180 + \lambda)$.

Now,

$$\begin{aligned} E(A_4 | a_1, a_2, a_3, a_4) &= \frac{2500}{250} E(X_4 | a_1, a_2, a_3) \\ &= \frac{2500}{250} \times 250 \times \frac{(\alpha + \frac{100a_1}{1500} + \frac{130a_2}{1700} + \frac{180a_3}{2000})}{(100 + 130 + 180 + \lambda)} \\ &= 2500 \times \frac{\alpha}{\lambda} \times \frac{\lambda}{(100 + 130 + 180 + \lambda)} + \frac{2500 \times (\frac{100a_1}{1500} + \frac{130a_2}{1700} + \frac{180a_3}{2000})}{(100 + 130 + 180)} \times \frac{(100 + 130 + 180)}{(100 + 130 + 180 + \lambda)} \\ &= 2500 \times \frac{\alpha}{\lambda} \times (1 - Z) + b \times Z \end{aligned}$$

Where,

$$Z = \frac{(100 + 130 + 180)}{(100 + 130 + 180 + \lambda)}$$

$$\text{And } b = \frac{2500 \times (\frac{100a_1}{1500} + \frac{130a_2}{1700} + \frac{180a_3}{2000})}{(100 + 130 + 180)}$$

[7]

ii) If the number of policies (i.e. the past experience) increases then Z becomes closer to 1 and hence more weight is placed on actual experience and less on the prior expectations. [1]

[8 Marks]

Solution 4:-

i) Let X denote the original claim amount

The reinsurance policy is such that the reinsurer pays X_R where X_R equals

$$\begin{aligned} X_R &= 0 && \text{if } X \leq 250 \\ &= X - 250 && \text{if } 250 < X \leq M \\ &= M - 250 && \text{if } X > M \end{aligned}$$

We also know that $E[X_R] = 500$

Thus we need to find M such that,

$$\int_{250}^M (x - 250)\lambda e^{-\lambda x} dx + \int_M^{\infty} (M - 250)\lambda e^{-\lambda x} dx = 500 \dots\dots\dots(i)$$

The LHS of the above equation can be written as

$$\begin{aligned} &[-xe^{-\lambda x}]_{250}^M + \int_{250}^M e^{-\lambda x} dx - 250[-e^{-\lambda x}]_{250}^M + [-(M-250)e^{-\lambda x}]_{M}^{\infty} \\ &= -M e^{-\lambda M} + 250 e^{-250\lambda} + [-e^{-\lambda x}/\lambda]_{250}^M + 250 e^{-\lambda M} - 250 e^{-250\lambda} + (M-250) e^{-\lambda M} \end{aligned}$$

Given that we know $1/\lambda = 1000$ the above equation becomes,

$$-M e^{-\lambda M} + 250 e^{-250\lambda} - 1000 e^{-\lambda M} + 1000 e^{-250\lambda} + 250 e^{-\lambda M} - 250 e^{-250\lambda} + M e^{-\lambda M} - 250 e^{-\lambda M}$$

Thus equation (i) becomes

$$500 = 1000 e^{-250\lambda} - 1000 e^{-\lambda M}$$

$$\text{Or } 500 = 1000 e^{-0.25} - 1000 e^{-0.001M}$$

$$\text{Or } e^{-0.001M} = (1000 e^{-0.25} - 500) / 1000$$

Thus,

$$M = -\text{LN}(e^{-0.25} - .5) / 0.001$$

$$\text{Or } M = 1.27725/0.001$$

$$\text{Or } M = 1277.25$$

[8]

ii) We need to find out the net average net claim amount paid by the insurer net of the deductible

At first we must find the size of the policy excess E such that

$$\int_0^E f(x) dx = 0.40$$

$$\int \alpha \lambda^\alpha / (x + \lambda)^{\alpha+1} dx = .40$$

$$\text{Or } 0.4 = [\lambda^\alpha / (x + \lambda)^\alpha]_0^E$$

$$= 1 - \lambda^\alpha / (E + \lambda)^\alpha$$

$$= 1 - (600 / (600 + E))^3$$

$$\text{Or } 600 / (600 + E) = .60^{1/3}$$

$$\text{Or } E = 600 / 0.8434 - 600 = 111.37$$

Now, the average net claim amount net of the deductible paid by the insurer is given by

$$E (X - E | X > E)$$

$$\text{Now, } \int_E^\infty (x - E) f(x) dx$$

$$= \int_E^\infty x \alpha \lambda^\alpha / (x + \lambda)^{\alpha+1} dx - \int_E^\infty E \alpha \lambda^\alpha / (x + \lambda)^{\alpha+1} dx$$

$$= [-x \lambda^\alpha / (x + \lambda)^\alpha]_E^\infty + \int \lambda^\alpha / (x + \lambda)^\alpha dx - E [-\lambda^\alpha / (x + \lambda)^\alpha]_E^\infty$$

$$= E \lambda^\alpha / (E + \lambda)^\alpha + [-\lambda^\alpha / ((\alpha-1) (x + \lambda)^{\alpha-1})]_E^\infty - E \lambda^\alpha / (E + \lambda)^\alpha$$

$$= \lambda^\alpha / ((\alpha-1) (E + \lambda)^{\alpha-1})$$

$$= 600^3 / (2 * (600 + 111.37)^2)$$

$$= 216000000 / 1012094.554$$

$$= 213.41$$

$$\text{Thus } E (X - E | X > E) = 213.41 / 0.6$$

$$= 355.69$$

[7]

[15 Marks]

Solution 5:-

i)

The cumulative claim amounts and developments factors for North, South and West are calculated in the tables below:

North			
Accident Year\Development Year	0	1	2
FY 13	50	90	155
FY 14	45	105	
FY 15	60		
DF		2.052632	1.722222

South			
Accident Year\Development Year	0	1	2
FY 13	30	85	135
FY 14	55	105	
FY 15	65		
DF		2.235294	1.588235

West			
Accident Year\Development Year	0	1	2
FY 13	55	115	150
FY 14	40	95	
FY 15	50		
DF		2.210526	1.304348

The ultimate claim amounts for different accident year and region are shown in the table below:

North			
Accident Year\Development Year	0	1	2
FY 13	50.0	90.0	155.0
FY 14	45.0	105.0	180.8
FY 15	60.0	123.2	212.1

South			
Accident Year\Development Year	0	1	2
FY 13	30.0	85.0	135.0
FY 14	55.0	105.0	166.8
FY 15	65.0	145.3	230.8

West			
Accident Year\Development Year	0	1	2
FY 13	55.0	115.0	150.0
FY 14	40.0	95.0	123.9
FY 15	50.0	110.5	144.2

[6]

All the above amounts are in Cr.

ii)

Let Y_{ij} represents aggregate (ultimate) claim amount for region i and year j (accident year),

P_{ij} represents the risk volume here it is number of policies for region i and year j

$$X_{ij} = Y_{ij}/P_{ij}$$

Hence X_{ij} is calculated in the given table (figure are in lacs).

Region	Xij		
	FY 13	FY 14	FY 15
North	620.0	565.1	707.0
South	450.0	490.5	824.1
West	555.6	476.6	379.4

Now using the tabular definitions we have calculated the below statistics.

Region	Pi	$\sum P_{ij}X_{ij}$	X_i	$\sum P_{ij}(X_{ij} - X_i)^2$	$\sum P_{ij}(X_{ij} - \bar{X})^2$	$P_i^*(1-P_i/P)$
North	87.0	54,793.9	629.8	315,215.1	801,967.9	59.0
South	92.0	53,252.6	578.8	2,448,339.3	2,500,525.7	60.7
West	91.0	41,807.8	459.4	500,636.5	1,332,136.1	60.3
Total	270.0	149,854.2		3,264,190.9	4,634,629.6	179.9

..Col 1 (0.5), Col 2 (1.0), Col 3 (0.5), Col 4 (1.0), Col 5 (1.0), Col 6 (0.5)

Now,

$$E(m(\theta)) = \bar{X} = 1,49,854.2/270 = 555.02$$

$$E(s^2(\theta)) = \frac{3,264,190.9}{3 \times (3-1)} = 544,031.81$$

$$P^* = 179.9 / (3 \times 3 - 1) = 22.49$$

$$Var(m(\theta)) = \frac{\frac{1}{3 \times 3 - 1} \times 4,634,629.6 - E(s^2(\theta))}{22.49} = 1,569.20$$

Now credibility factor for South (Z2) can be calculated as given below:

$$Z_2 = \frac{\sum_j P_{ij}}{\sum_j P_{ij} + \frac{E(s^2(\theta))}{Var(m(\theta))}} = 0.20971$$

Hence credibility premium per each policy can be calculated as

$$= 0.20971 \times \bar{X}_2 + (1 - 0.20971) \times E(m(\theta)) = 560.01$$

Hence the ultimate claim for each accident year FY 16 and FY 17 is given by

$$\text{For FY 16, ultimate claim (in Cr.)} = 35 \times 560.01 / 100 = 196$$

$$\text{For FY 17, ultimate claim (in Cr.)} = 40 \times 560.01 / 100 = 224$$

[9]

iii)

Assumptions:

For each region i , the distribution of x_{ij} depends on a parameter θ_{ij} whose value is the same for each j but is unknown.

$\{X_{ij} | \theta_i\}$ are independent random variables.

$\{\theta_i\}$ are i.i.d random variables.

For i not equal to k , the pairs (X_{ij}, θ_i) and (X_{km}, θ_k) are independent random variables.

There exist functions $m(\cdot)$ and $s_2(\cdot)$ such that:

$$m(\theta_i) = E(X_{ij} | \theta) \quad s_2(\theta_i) = P_{ij} \text{var}(X_{ij} | \theta)$$

[1]

iv)

The reported claims for South region for various years are given in the below table:

	Reporting Year							
Accident Year	FY 13	FY 14	FY 15	FY 16	FY 17	FY 18	FY 19	Ultimate liability
FY 13	30.0	55.0	50.0					135.0
FY 14		55.0	50.0	61.8				166.8

FY 15			65.0	80.3	85.5			230.8
FY 16				55.2	68.2	72.6		196.0
FY 17					63.1	77.9	83.0	224.0
Total	30.0	110.0	165.0	197.3	216.8	150.5	83.0	

The reported claim in FY 16 for Accident Year FY 16 is calculated as given below

$$= 196 / (2.235294 * 1.588235) = 55.2$$

The reported claim in FY 17 for Accident Year FY 16 is calculated as given below

$$= 55.2 * 2.235294 - 55.2 = 68.2$$

Similarly other numbers are calculated.

[4]

[20 Marks]

Solution 6:-

i)

Let assume that the Poisson rate for car, truck, two wheelers and buses insurance are $\lambda_1, \lambda_2, \lambda_3$ and λ_4 .

The likelihood function for car insurance is given by:

$$L = e^{-5\lambda_1} \frac{(5\lambda_1)^{65}}{65!}$$

Taking log we get

$$l = \log L = -5\lambda_1 + 65 \log(5\lambda_1) + C, \text{ where } C \text{ is constant}$$

Differentiating we get,

$$\frac{dl}{d\lambda_1} = -5 + \frac{65}{\lambda_1}, \frac{d^2l}{d\lambda_1^2} = -\frac{65}{\lambda_1^2} < 0$$

Hence, $\lambda_1 = \frac{65}{5} = 13$ and it has a maximum value.

Similarly, $\lambda_2 = \frac{45}{3} = 15$

$$\lambda_3 = \frac{85}{10} = 8.5$$

$$\lambda_4 = \frac{75}{6} = 12.5 \quad [4]$$

ii)

We have to check that the mean claim rates for difference insurance type are same or not

$$\text{ie, } \lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = \lambda$$

$$\text{The mle of } \hat{\lambda} \text{ is given by } \hat{\lambda} = \frac{(65 + 45 + 85 + 75)}{(5 + 3 + 10 + 6)} = 11.25$$

The difference in scaled deviance will follow chi-square distribution with $(4-1) = 3$ degrees of freedom.

The scaled deviance can be calculated using the below equation:

$$\begin{aligned} & 2(\text{Log}L_1 + \text{Log}L_2 + \text{Log}L_3 + \text{Log}L_4 - \text{Log}L) \\ &= 2[(65\text{Log}\lambda_1 - 5\lambda_1) + (45\text{Log}\lambda_2 - 3\lambda_2) + (85\text{Log}\lambda_3 - 10\lambda_3) + (75\text{Log}\lambda_4 - 6\lambda_4) - (270\text{Log}\hat{\lambda} - 24\hat{\lambda})] \\ & \text{(Other items cancelled out in the difference)} \\ &= 2[(65\text{Log}13 + 45\text{Log}15 + 85\text{Log}8.5 + 75\text{Log}12.5 - 270\text{Log}11.25 - (65 + 45 + 85 + 75 - 270))] \\ &= 2*6.4198 = 12.8397 \end{aligned}$$

The critical value at upper 5% level for chi-square distribution with 3 degrees of freedom is 7.815. The calculated statistic is greater than the critical value. Hence mean rates are different for different insurance types. [5]

[9 Marks]

Solution 7:-

i)

Let R denote tasting a RaisinRich chocolate and N denote tasting a NutRich chocolate

The different possibilities are

$$D_1(R) = 7R \text{ and } D_1(N) = 2R$$

$$D_2(R) = 7R \text{ and } D_2(N) = 7R$$

$$D_3(R) = 2R \text{ and } D_3(N) = 2R$$

$$D_4(R) = 2R \text{ and } D_4(N) = 7R$$

Where 7R refers to selection of RaisinRich box and 2R refers to NutRich box selection.

Under 7R, Prob (R) = .7 and Prob (N) = .3

Under 2R, Prob (R) = .2 and Prob (N) = .8

The expected loss to the actuarial student is either 75 or 90 depending on whether he selects the chocolate box correctly or incorrectly respectively.

The risk function under for each decision function and chocolate box is as given below:

$$R(D_1, 7R) = P(R|7R) * 75 + P(N|7R) * 90 = .7 * 75 + .3 * 90 = 79.5$$

$$R(D_2, 7R) = P(R|7R) * 75 + P(N|7R) * 75 = .7 * 75 + .3 * 75 = 75$$

$$R(D_3, 7R) = P(R|7R) * 90 + P(N|7R) * 90 = .7 * 90 + .3 * 90 = 90$$

$$R(D_4, 7R) = P(R|7R) * 90 + P(N|7R) * 75 = .7 * 90 + .3 * 75 = 85.5$$

$$R(D_1, 2R) = P(R|2R) * 90 + P(N|2R) * 75 = .2 * 90 + .8 * 75 = 78$$

$$R(D_2, 2R) = P(R|2R) * 90 + P(N|2R) * 90 = .2 * 90 + .8 * 90 = 90$$

$$R(D_3, 2R) = P(R|2R) * 75 + P(N|2R) * 75 = .2 * 75 + .8 * 75 = 75$$

$$R(D_4, 2R) = P(R|2R) * 75 + P(N|2R) * 90 = .2 * 75 + .8 * 90 = 87$$

The risk function represented in the form of a matrix is

Risk function matrix	D1	D2	D3	D4
7R	79.5	75	90	85.5
2R	78	90	75	87

[8]

ii) Under the Bayes Criterion

Probability (7R) = .4

Probability (2R) = .6

$$\text{Expected loss (D1)} = 79.5 * .4 + 78 * .6 = 78.6$$

$$\text{Expected loss (D2)} = 75 * .4 + 90 * .6 = 84$$

$$\text{Expected loss (D3)} = 90 * .4 + 75 * .6 = 81$$

$$\text{Expected loss (D4)} = 85.5 * .4 + 87 * .6 = 86.4$$

Hence under the Bayes Criterion, the optimum strategy is to choose strategy D1 for which the expected loss is the minimum. [3]

[11 Marks]

Solution 8:-

i)

If θ be the population parameter to be estimated and let $g(X)$ be used as an estimator of θ

Then two loss functions apart from quadratic loss functions are given below:

a) Absolute error loss which is defined as

$$L(g(x), \theta) = |g(x) - \theta|$$

b) All or nothing loss or 0/1 loss function which is defined as

$$L(g(x), \theta) = 0 \text{ if } g(x) = \theta$$

$$= 1 \text{ if } g(x) \neq \theta \quad [2]$$

ii) Quadratic loss function is defined as $L(g(x), \theta) = (g(x) - \theta)^2$

$$\text{Expected Posterior loss} = \int (g(x) - \theta)^2 \cdot f(\theta|x) d\theta$$

Here the integral is assumed to cover the range $(-\infty, \infty)$

$$\text{Or, } d/dg(\text{EPL}) = 2 \int (g(x) - \theta) f(\theta|x) d\theta \quad (\text{by using the Leibnitz rule given in the Actuarial tables})$$

To find out the minimum we need to equate the above equation to zero

Hence,

$$\int g(x) f(\theta|x) d\theta = \int \theta f(\theta|x) d\theta$$

$$\text{Or } g(x) \int f(\theta|x) d\theta = \int \theta f(\theta|x) d\theta$$

$$\text{However, } \int f(\theta|x) d\theta = 1$$

$$\text{Hence, } g(x) = \int \theta f(\theta|x) d\theta = E(\theta|x) \dots\dots\dots (i) \text{ which minimises EPL}$$

As Differentiating EPL twice gives

$$2 \int f(\theta|x) d\theta = 2 > \text{zero} \dots\dots\dots (ii)$$

Thus equation (i) and (ii) show that the Bayesian Estimator under quadratic loss is the mean of the posterior distribution. [3]

[5 Marks]

Solution 9:-

i) The discrete uniform distribution would have a probability of 0.25 for each of the values 0, 1, 2 and 3.

The highest probability in our target distribution is 0.75 (corresponding to a value of 0).

So, to apply the acceptance/rejection method, we would accept all 0's from the discrete uniform distribution, but reject other values with the appropriate probabilities to obtain the correct proportions of each number.

The acceptance/rejection decision is made as follows, using pseudo-random numbers U from a uniform distribution on (0,1) :

Discrete uniform value	Accept if ...
0	Always
1	$U < 0.2$ (ie 0.15 / 0.75)
2	$U < 0.107$ (ie 0.08 / 0.75)
3	$U < 0.027$ (ie 0.02 / 0.75)

[4]

(ii) Proportion accepted

The proportion of values accepted by this method would be:

$$0.25 + 0.25 * 0.2 + 0.25 * 0.107 + 0.25 * 0.027 = 0.33333 \text{ ie } 33.33\%$$

This means that we need to generate more than 3 discrete random values to generate each value from the target distribution. So this method is not very efficient.

[3]

(iii) Alternative method

Use the inverse transformation method: generate values from a continuous uniform distribution on the range (0,1) . Then return the values shown in the table below, based on the range in which the uniform value fell.

Uniform value	Return
0.00 – 0.75	0
0.75 – 0.90	1
0.90 – 0.98	2
0.98 – 1.00	3

[3]

[10 Marks]**Solution 10:-**

i) When aggregate claims are a compound Poisson process, the adjustment coefficient is defined in terms of the Poisson parameter (λ), the moment generating function of individual claim amounts ($MX(r)$) and the premium income per unit time (c).

The adjustment coefficient, denoted R , is defined to be the unique positive root of:

$$\lambda MX(r) - \lambda - cr = 0$$

b. The adjustment coefficient equation is:

$$\lambda + cr = \lambda E(\exp(rX))$$

[1]

ii) The average claim size is:

$$E(X) = 0.1 * 15000 + 0.9 * 2000 = \text{Rs } 3300$$

So the annual risk premium is:

$$E(X) = 0.03 * 3300 = \text{Rs } 99.$$

So the annual office premium is $c = 1.3 * 99 + 30 = 158.70$

So the adjustment coefficient equation is:

$$0.03 + 158.70r = 0.03(0.9\exp(2000r) + 0.1\exp(15000r))$$

Simplifying by dividing by 0.03 and writing $R = 1,000r$ gives:

$$1 + 5.29R = 0.9e^{2R} + 0.1e^{15R}$$

Expanding the RHS as a series to get a first approximation:

$$\begin{aligned} 1 + 5.29R &= 0.9(1 + 2R + 2R^2 + \dots) + 0.1(1 + 15R + 112.5R^2 + \dots) \\ &= 1 + 3.3R + 13.05R^2 + \dots \end{aligned}$$

Ignoring higher orders of R , R approximately equals to 0.1525.

Evaluating $f(R) = 1 + 5.29R - 0.9e^{2R} - 0.1e^{15R}$ for different values of R we get,

$$f(0.15) = 0.37, f(0.1) = .018, f(.09) = -.013, f(.094) = -.0015, f(.095) = .00156 \text{ \& } f(.0945) = .000011$$

So R lies between 0.094 and 0.0945. So the value of the adjustment coefficient correct

To 2SF is 0.000094.

[9]

[10 Marks]
